Self-Interference Incoherent Digital Holography for Holographic Fluorescence Microscopy

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Abstract: We present recent progress in the development of self-interference incoherent digital holography for holographic fluorescence microscopy, including aberration compensation of fluorescent holographic image and 3D differential holography.
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1. Introduction

Fluorescence microscopy is an indispensable imaging tool in modern biomedical research. Holography is well-known to have many interesting and useful imaging capabilities. The requirement of coherent illumination has all but precluded holography as a means for fluorescence imaging, which is inherently incoherent. Recent developments in digital holography, however, including our self-interference incoherent digital holography (SIDH) [1,2], provide highly effective and versatile capabilities for 3D holographic imaging with incoherent light, that can remove the barrier between fluorescence and holography.

In order to break the coherence barrier and to expand its applications, there have been numerous efforts to achieve holographic recording using incoherent light. One approach is to use short-coherence light source in an interferometer with closely-matched path lengths [3]. Or, one can use a spatial-filtered copy of the object field as the reference, such as in diffraction phase microscopy and related techniques [4]. Another approach is based on self-interference of two copies of the object field with a radial shear, i.e. differential curvature, between them. The radial shear may be achieved by conoscopic birefringence [5], triangular interferometer [6], or by a spatial light modulator [7-9]. Optical scanning holography was the first successful method to achieve holographic recording of fluorescence [10].

2. Methods and results

We have constructed a holographic camera based on the SIDH principle (Fig. 1). It consists of a beam splitter and a pair of mirrors with different curvatures. A plane wave originating from a source point is split into two copies and different curvatures are imparted on them before superposing on the camera, resulting in a Fresnel lens-like concentric interference pattern, as shown in inset. Given the spatial incoherence of the object illumination, these patterns from multiple source points add incoherently, quickly building up large dc background. In order to eliminate the dc and twin image terms, one can either use in-line phase shifting multi-frame acquisition or off-axis single-shot acquisition. The holographic camera can be attached to various conventional optical instruments with minimal modification, such as optical microscopes, telescopes, and photographic cameras. Figure 2 is a holographic image of the moon acquired with the holographic camera in telescopic mode, which represents the first ever recording of an astronomical object using a holographic process of any kind.

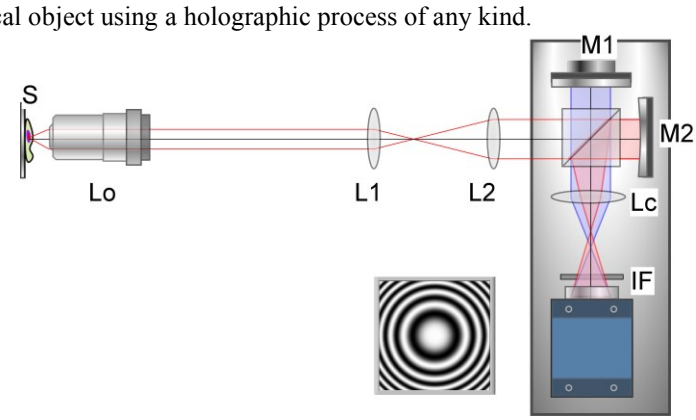

Fig. 1: Schematic of holographic fluorescence microscope (HFM). M1: piezo-mounted plane mirror; M2: curved mirror. Inset: Fresnel lens-like interference pattern formed from a source point.

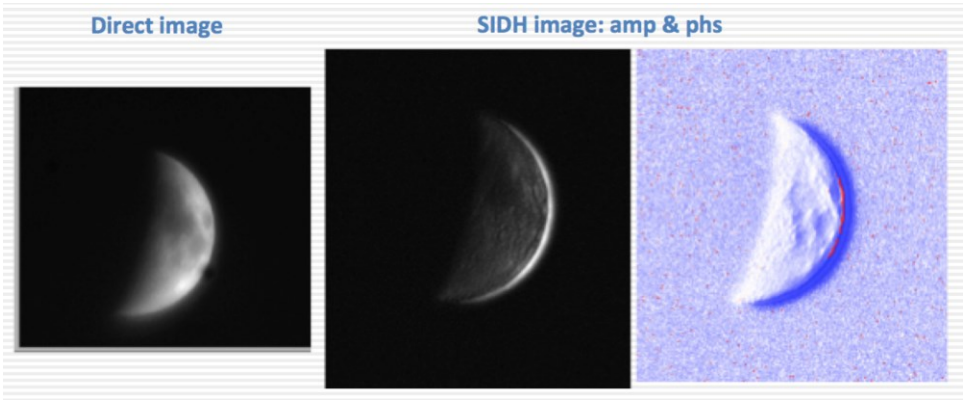

Fig. 2: Holographic image recorded with a SIDH-based holographic camera.

In Fig. 3a) and b), a full-field holographic image of a fluorescent object is taken, consisting of a resolution target with a fluorescent plate behind it and illuminated with an ultraviolet lamp. An aberrator – a broken piece of curved glass –placed in the optical path causes the holographic image to be significantly distorted. SIDH is a holographic process and, as such, it records the complete amplitude and phase profile of the optical field. In particular, the hologram of a point source object is the complex point spread function (PSF) of the SIDH system. This is in contrast to the intensity-only PSF of conventional fluorescence microscopy systems. We demonstrate that the guide star hologram can be used to measure the aberration of the optical system, Fig. 3c), and to compensate the hologram of the full-field fluorescent object, shown in Fig. 3d) and e). The process represents deterministic deconvolution by complex PSF, as opposed to non-deterministic deconvolution by intensity PSF. Thus the SIDH image can be enhanced by correlation with a guide star hologram. Also significant to note is that weak image signals buried under noise can be amplified by using a brighter guide star, [1].

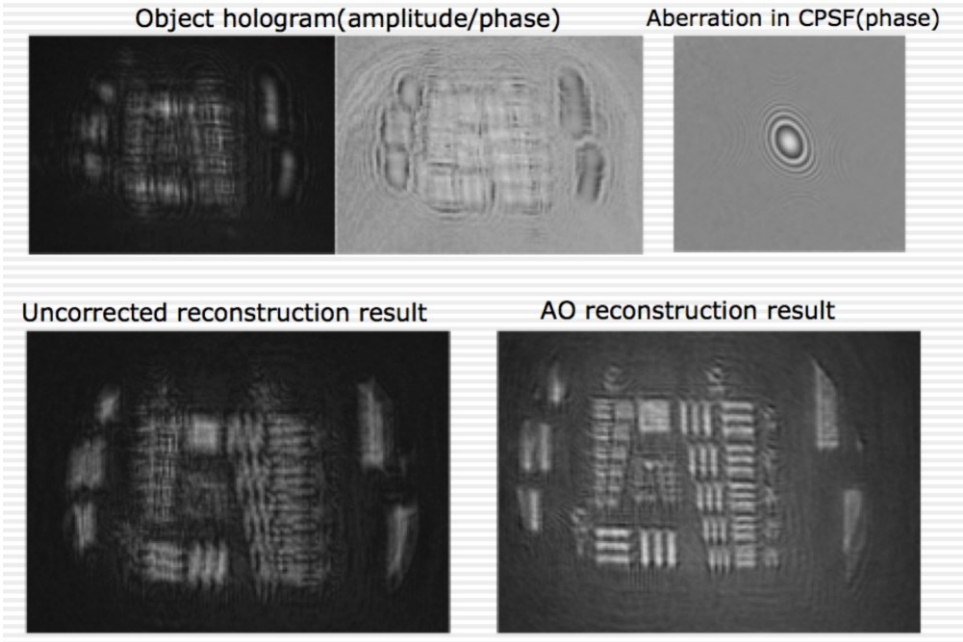

Fig. 3: Complex hologram acquired by SIDH of a fluorescent object, consisting of a resolution target with a fluorescent plate behind it and illuminated with an ultraviolet lamp. An aberrator, broken piece of glass, causes distortion of the hologram, as measured by SIDH of a point object. Correlation of the guide star hologram with the full-field hologram removes most of the effect of the aberration.

In differential SIDH, two holograms are acquired by SIDH while some structures or components change in the object scene. A single subtraction of one complex hologram from the other results in a difference hologram that contains the three-dimensional dynamics of the object space volume as shown in Fig. 4. Differential SIDH provides an efficient means to obtain 3D dynamic changes of an object volume. Differential fluorescence holographic

microscopy of cellular processes will be presented, such as in mitosis or motility of cells tagged with fluorescent proteins.

Fig. 4: Images reconstructed from SIDH holograms taken before and after the knight's move on a chess board, illuminated with daylight from an office window. The difference hologram contains full 3D information of the changed scene and absence of any static information.

3. Conclusion

This paper presents recent progress in the development of SIDH for applications in holographic fluorescence microscopy, including differential fluorescence holography and aberration compensation of fluorescence microscopy.

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